

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

&

The Patent Rules, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

TITLE OF THE INVENTION

“Intelligent AI-Driven Job Recommendation and Resume Matching System with Multi-Domain Skill Analysis, Contextual Compatibility Scoring, and Integrated Voice-Enabled Career Assistant”

Applicant(s)

NAME	NATIONALITY	ADDRESS
1. Satej Mahendra Rajeshirke	Indian	Student, Department of Computer Engineering, Jaihind College of Engineering, Kuran, Junnar, Pune-410511
2. Jaya Mahendra Rajguru	Indian	Student, Department of Computer Engineering, Jaihind College of Engineering, Kuran, Junnar, Pune-410511
3. Eshwar Mahadeo Sambherao	Indian	Student, Department of Computer Engineering, Jaihind College of Engineering, Kuran, Junnar, Pune-410511
4. Prof Karad Vandana Ambadas	Indian	Assistant Professor, Department of Computer Engineering, Jaihind

		College of Engineering, Kuran, Junnar, Pune-410511
5. Prof Bhosale Sachin Bajirao	Indian	Assistant Professor, Department of Computer Engineering, Jaihind College of Engineering, Kuran, Junnar, Pune-410511
6. Dr. Gunjal Shubhangi Dharmendra	Indian	Associate Professor, Department of Computer Engineering, Jaihind College of Engineering, Kuran, Junnar, Pune-410511
7. Dr. Khatri Anand Ashok	Indian	Associate Professor, Department of Computer Engineering, Jaihind College of Engineering, Kuran, Junnar, Pune-410511
8. Prof Bangar Abhishek Popat	Indian	Assistant Professor, Department of Computer Engineering, Jaihind College of Engineering, Kuran, Junnar, Pune-410511

The following specification particularly describes the nature of the invention and the manner in which it is performed:

FIELD OF THE INVENTION

[001] The invention relates to the domain of artificial intelligence-enabled recruitment technologies, particularly systems designed for automated resume analysis, intelligent job-candidate matching, multi-domain skill extraction, contextual compatibility scoring, and voice-enabled career assistance. The invention further pertains to integrated full-stack architectures that combine frontend interfaces, backend data management, and AI microservices into a unified platform to deliver real-time, personalized job recommendations and career insights.

BACKGROUND OF THE INVENTION

[002] The field of automated recruitment technologies has evolved substantially over the last decade; however, existing systems continue to rely heavily on conventional keyword-matching mechanisms that fail to capture the deeper semantics of candidate capabilities. Traditional resume screening tools extract only surface-level information without understanding the relationship between skills, experiences, and job responsibilities. As a result, hiring decisions are often made using incomplete or context-deficient data, leading to inefficiencies for recruiters and job seekers alike.

[003] Most commercial job portals and applicant tracking systems employ narrow NLP models that treat resumes as static text bodies, ignoring the contextual meaning of each skill and the relevance of those skills in specific job domains. These limitations prevent the accurate mapping of a candidate's actual profile to the dynamic needs of modern industries, where skill relationships, proficiency levels, and domain-specific competencies are critical.

[004] Another challenge with existing solutions is the lack of real-time analysis. Many AI-based systems process data in batches, leading to delays between resume submission and feedback generation. This time lag creates poor user engagement and prevents candidates from receiving instant recommendations or actionable guidance while navigating the platform.

[005] Manual resume screening continues to dominate the recruitment process for many organizations, consuming significant time and human resources. Recruiters must validate skills, interpret experience levels, and determine overall suitability without the support of consistent and intelligent automation. Human screening also introduces subjectivity, inconsistency, and potential bias, thereby impacting the fairness and reliability of candidate evaluation.

[006] Existing resume parsers typically fail to validate whether a candidate's claimed skills are genuinely present in the context of their experience. For example, a skill may appear only as a keyword without any evidence of practical usage. Conventional systems lack the ability to confirm contextual relevance, leading to inaccurate extraction and misleading candidate profiles.

[007] Job-candidate matching algorithms available today are generally rigid, rule-based, and dependent on fixed scoring systems. These systems cannot adapt to unique job requirements, evolving industry standards, or the relative importance of different skills across multiple domains. As industries grow more multidimensional, the need for an adaptive, AI-driven matching framework has become critical.

[008] Additionally, no widely used recruitment platform provides an integrated voice-enabled career assistant capable of interpreting spoken commands, offering insights, and guiding users through the job search or evaluation process. Voice interfaces are typically rudimentary and disconnected from the underlying AI-powered analysis, preventing them from performing intelligent, career-oriented functions.

[009] A further limitation in prior systems is the fragmented architecture where the frontend, backend, and AI services operate in isolated silos. This fragmentation causes delays, inconsistent user experiences, and inability to synchronize analysis results in real time. Without unified communication and synchronized data flow, it becomes impossible to deliver instantaneous job recommendations or dynamic career guidance.

[010] Therefore, there exists a need for an integrated, AI-powered recruitment platform capable of real-time resume processing, contextual skill validation, domain-aware compatibility scoring, and seamless voice-assisted interaction. Such a system must unify all components—frontend, backend, and AI processing—into a synchronized architecture to deliver accurate, fast, and personalized job recommendations for both candidates and recruiters.

SUMMARY OF THE INVENTION

[011] The invention provides an intelligent, AI-driven recruitment system designed to overcome the limitations of existing job-matching technologies by integrating advanced natural language processing, multi-domain skill analysis, contextual resume interpretation, and dynamic compatibility scoring within a unified full-stack architecture. The system enables real-time understanding of a candidate's true capabilities and aligns them accurately with job requirements across diverse industries.

[012] At the core of the invention is a sophisticated AI microservice that performs multi-layered resume parsing, semantic feature extraction, and contextual skill validation. Unlike conventional keyword-based methods, the system evaluates each skill in relation to surrounding text, job responsibilities, and demonstrated experience, ensuring a deeper and more meaningful interpretation of the candidate's profile.

[013] The system incorporates a comprehensive hierarchical skill database spanning multiple industries, allowing it to categorize, validate, and score capabilities with high precision. This domain-aware structure empowers the matching engine to weight different skills appropriately depending on the specific role being analyzed, thereby producing a more accurate and role-aligned compatibility score.

[014] A key component of the invention is the real-time job recommendation engine that computes compatibility scores using dynamically adjustable weighting factors for skills, experience, education, and domain relevance. The engine continuously adapts its analysis based on user interactions, market

changes, and evolving industry trends, resulting in progressively refined and personalized recommendations.

[015] The invention further includes an intuitive frontend interface that delivers instant feedback and visualizations, such as match percentages, skill alignment graphs, and experience-context summaries. The frontend seamlessly interacts with the backend and AI microservices through synchronized communication channels, ensuring instant updates whenever a resume is processed or user preferences are modified.

[016] The backend server manages user authentication, job listings, application workflows, session states, and secure data storage. It maintains structured relationships between candidates, recruiters, resumes, job postings, and AI-generated insights. This ensures consistency, reliability, and integrity of data across all interactions and system components.

[017] Another novel aspect of the invention is its integrated voice-enabled career assistant, which allows users to navigate the platform, request job insights, receive career guidance, and execute application-related actions through spoken commands. The voice assistant employs advanced speech recognition and natural language understanding to interpret commands accurately and respond contextually, enhancing accessibility and user engagement.

[018] The invention's unified architecture enables seamless synchronization between frontend, backend, AI microservices, and voice-processing components. This real-time data flow ensures that every user action—whether uploading a resume, viewing job details, or issuing a voice command—triggers immediate processing and feedback, eliminating delays prevalent in prior systems.

[019] The invention thus provides a holistic, highly adaptable, and technologically advanced solution for modern recruitment challenges. By integrating AI-driven skill analysis, contextual interpretation, dynamic matching, voice interaction, and full-stack synchronization, the system significantly

enhances the efficiency, accuracy, and user experience for both job seekers and recruiters.

BRIEF DESCRIPTION OF THE DRAWINGS

[020] The drawings accompanying the present specification illustrate the functional flow, architectural structure, and interactive processes implemented by the invention. These drawings serve to enhance understanding of the system's core components, data movement, and operational mechanisms.

[021] One drawing depicts the **Resume Processing Flowchart**, illustrating the step-by-step procedure through which an uploaded resume is received, pre-processed, semantically analyzed, and converted into structured data by the AI microservice. It further outlines the pathways for skill extraction, contextual validation, and experience interpretation.

[022] Another drawing presents the **Voice Interaction Sequence Diagram**, showing how spoken commands are captured through the user interface, transmitted to the voice-processing module, interpreted using speech recognition and natural language understanding, and subsequently executed through backend and AI services. The diagram highlights the bidirectional communication between user, system components, and generated responses.

[023] A separate drawing demonstrates the **Real-Time Matching Process**, outlining the flow of data from resume extraction to compatibility-score computation. This diagram illustrates how the system dynamically adjusts weighting factors, processes job requirements, and synchronizes match results for instant display on the frontend.

[024] An additional drawing represents the **Unified System Architecture Diagram**, providing a high-level view of interactions between the frontend interface, backend server, AI microservice, voice assistant, and database components. This figure emphasizes the real-time synchronization and coordinated operation of all modules in the invention.

[025] The drawings collectively support a clearer interpretation of the invention's mechanisms and highlight the novel integration of AI processing, voice assistance, domain-aware analysis, and real-time frontend-backend coordination.

DETAILED DESCRIPTION OF THE INVENTION

System Overview

[026] The invention provides an integrated, intelligent recruitment and career-guidance platform that combines artificial intelligence, natural language processing, multi-domain skill analysis, dynamic job-matching algorithms, and voice-assisted interaction within a unified full-stack architectural framework. The system is designed to process resumes in real time, extract structured and contextually validated information, compute compatibility scores aligned with specific job requirements, and deliver personalized job recommendations through both visual and voice interfaces. The invention ensures synchronized operation of all components through coordinated data flow mechanisms that link the frontend interface, backend server, AI microservices, and voice-processing units.

[027] The architecture of the invention comprises four principal components: a responsive frontend user interface, an AI processing microservice for resume interpretation and job matching, a backend server with structured database management, and an integrated voice assistant capable of speech recognition and contextual interpretation. These components operate collaboratively through RESTful APIs, WebSocket channels, and synchronized event triggers, ensuring immediate feedback and continuous data coherence across all modules.

Frontend Interface

[028] The frontend system serves as the primary interaction layer for candidates and recruiters. It is implemented using a dynamic web framework that supports adaptive rendering, responsive layouts, and interactive components. The user

interface includes sections for resume upload, real-time skill visualization, job recommendation display, application tracking, and dashboard-based analytics. When a resume is uploaded, the frontend immediately transfers the file to the AI microservice and displays live processing indicators to enhance transparency and user engagement.

[029] The job recommendation module within the frontend interface presents AI-computed compatibility scores in a structured and intuitive manner. Job postings are listed along with detailed breakdowns of skill alignment, experience match, and education relevance. Visual aids such as graphs, color-coded bars, competency matrices, and gap-analysis indicators allow users to understand the reasoning behind each recommendation. The frontend also includes interfaces for recruiters to review candidate profiles enriched with AI-generated insights.

[030] A unique element of the frontend is its integrated voice-assistant overlay, positioned persistently throughout the interface. This voice-control module provides speech waveform visuals, recognition feedback, and contextual prompts. Users can issue commands such as “show recommended jobs,” “analyze my skills,” “filter jobs by domain,” or “explain match score,” enabling a hands-free and accessible experience.

AI Processing Microservice

[031] The AI microservice is responsible for performing multi-layered semantic analysis of resumes. It executes optical character recognition (if required), text normalization, sentence segmentation, and token-level parsing. The system subsequently applies transformer-based language models, domain-trained NLP pipelines, and hierarchical classifiers to extract relevant features such as skill sets, project experience, educational qualifications, certifications, job history, and inferred capabilities.

[032] The invention utilizes a comprehensive multi-domain skill database that categorizes competencies across technical, managerial, analytical, and soft-skill domains. The AI microservice validates the presence of each skill not

merely by keyword occurrence but by analyzing contextual indicators such as verb-skill relationships, project descriptions, responsibility patterns, and achievement statements. This contextual skill validation allows the system to distinguish between genuinely practiced skills and those mentioned superficially.

[033] The experience analyzer determines work durations, identifies career progression stages, and estimates depth of proficiency based on textual patterns. For example, the system can infer seniority levels, detect transitions between industries, evaluate consistency of role responsibilities, and measure alignment between job titles and described duties. The education analyzer standardizes degree names, specializations, and academic levels to conform to a uniform comparison framework.

Matching Engine and Scoring Logic

[034] The invention's matching engine computes compatibility scores based on a multi-factor weighted model. Each job requirement is decoded into required and preferred skill sets, minimum experience levels, educational thresholds, and domain-specific competencies. The system applies adaptive weighting factors that adjust dynamically depending on job nature, market trends, and skill importance. The matching logic compares extracted resume features to job parameters and assigns scores using contextual similarity functions and domain-based weight adjustments.

[035] The matching engine performs continuous recalibration as users browse job postings, update preferences, or interact with system elements. This progressive learning mechanism captures user behavior to refine recommendations over time. For instance, if a user frequently selects data-science roles, the system learns to emphasize analytical and machine-learning skills in subsequent recommendations, even when multiple domains are present in the resume.

Backend Server and Database System

[036] The backend server manages authentication, job posting workflows, resume storage, application tracking, and AI-result persistence. It maintains relational tables mapping users, resumes, job listings, extracted attributes, and computed match analyses. The backend ensures that each update from the AI microservice is recorded instantly and is retrievable by the frontend without delay. Secure role-based access control allows candidates, recruiters, and administrators to operate within defined privileges.

[037] The backend also orchestrates synchronous communication among all components. When the AI microservice generates structured outputs, the backend stores them while simultaneously sending update notifications to the frontend. When a voice command is processed, the backend executes the requested action—such as retrieving job recommendations or updating user preferences—and returns the results to the voice engine and user interface in real time.

Voice Assistant Module

[038] The voice assistant is built on advanced speech-recognition technology capable of interpreting natural-language commands with high accuracy across varied accents and environments. The system employs noise reduction, intent classification, and entity extraction to convert spoken language into actionable intents. The assistant understands job-search terminology, navigational phrases, and request-based queries such as “find jobs that match my experience” or “explain why this job is a good fit.”

[039] The voice assistant not only executes commands but also provides intelligent explanations. When a user asks, “Why is this job recommended for me?”, the system retrieves compatibility parameters and verbally summarizes skill alignment, experience fit, and education relevance. This interactive conversational capability enhances accessibility for users who prefer voice-based navigation or face difficulties with manual interaction.

Real-Time Data Flow and System Synchronization

[040] A key feature of the invention is its unified data flow architecture. The system maintains real-time synchronization between the frontend interface, backend services, AI analysis modules, and voice processing units using WebSockets and structured API calls. Whenever a resume is uploaded, processed, or modified, all connected modules receive instantaneous updates, enabling seamless user experience with no perceptible delays.

[041] This synchronous design ensures that job recommendations, skill graphs, voice responses, and database entries remain fully consistent. Any action taken in one component—such as issuing a voice command, editing a profile, or viewing job analysis—reflects immediately across all others, eliminating fragmentation and latency seen in conventional recruitment systems.

[042] The invention described herein presents an advanced, fully integrated AI-driven recruitment and career-assistance platform that significantly enhances the accuracy, speed, and personalization of job matching processes. Through the combined use of semantic resume analysis, multi-domain skill validation, adaptive compatibility scoring, and voice-enabled interaction, the system provides a holistic solution that overcomes the deficiencies of existing recruitment technologies. The unified real-time architecture ensures seamless communication between all modules, offering a responsive and contextually intelligent experience for both candidates and recruiters.

[043] The invention further establishes a foundation for scalable and extensible applications, as its modular design allows additional AI capabilities to be incorporated without disrupting existing workflows. Future improvements may include integration of predictive analytics for career forecasting, AI-driven gap-training recommendations, employer behavioral insight modeling, and multilingual voice-interaction frameworks to support broader accessibility. The system can also be expanded to include cross-platform mobile assistants, AR/VR-enabled interview simulations, and federated learning models that enhance privacy-preserving AI training across distributed datasets.

[044] As industries continue to evolve and new job roles emerge, the invention is well-positioned to adapt dynamically by updating domain-specific weightings, enhancing skill taxonomies, and incorporating industry-specific AI models. The system's ability to continuously learn from user interactions ensures long-term improvement in matching precision and recommendation relevance. This adaptability establishes the invention as a forward-looking solution for addressing the future demands of employers, job seekers, educational institutions, and workforce-planning agencies.

[045] Thus, the invention provides a technologically advanced, future-ready recruitment ecosystem that not only enhances current hiring practices but also enables strategic workforce development. The combination of AI-powered contextual analysis, intelligent scoring, real-time synchronization, and interactive voice assistance positions the system as a transformative innovation in the field of automated recruitment. While specific embodiments and architectural variations have been described, it will be understood that modifications, enhancements, and alternative implementations can be incorporated without departing from the scope of the invention as claimed.

We Claims

1. A computer-implemented intelligent job recommendation system comprising a frontend interface, an AI-based resume processing microservice, a backend server, and a voice-enabled assistant, wherein the system processes resumes in real time, extracts structured information, and generates job-candidate compatibility scores using contextual and multi-domain skill analysis.
2. The system as claimed in claim 1, wherein the AI microservice performs multi-layered natural language processing to extract skills, experience, education, and inferred competencies through semantic parsing, contextual validation, and domain-specific classification.
3. The system as claimed in claim 1, wherein the matching engine computes compatibility scores using a weighted model that dynamically adjusts based on skill relevance, experience alignment, education suitability, and industry-specific priority factors.
4. The system as claimed in claim 1, wherein the frontend interface displays real-time visualizations including skill graphs, competency matrices, compatibility indicators, and job-alignment insights derived from AI analysis.
5. The system as claimed in claim 1, wherein the backend server maintains synchronized communication between the frontend, AI microservices, and voice assistant through REST APIs and real-time data channels to ensure consistent updates across all modules.
6. The system as claimed in claim 1, wherein the voice-enabled assistant interprets spoken commands using speech recognition and natural language understanding to retrieve job recommendations, provide career insights, execute navigation actions, and verbally explain compatibility scores.
7. The system as claimed in claim 1, wherein the multi-domain skill database categorizes skills across diverse industries and enables the AI microservice to validate skill relevance based on contextual evidence within the resume text.
8. The system as claimed in claim 1, wherein the resume processing pipeline includes real-time extraction, validation, and transformation of unstructured resume content into structured data formats for immediate matching against job listings.

9. The system as claimed in claim 1, wherein the unified system architecture maintains real-time synchronization among all components such that any action performed by the user triggers instantaneous updates of job recommendations, skill analysis, and voice assistant responses.
10. The system as claimed in claim 1, wherein the system progressively adapts its job recommendation outputs based on user interactions, historical behavior patterns, and evolving industry trends to refine compatibility scoring over time.

Dated this 01st day of December 2025

Applicant(s)

Satej Mahendra Rajeshirke et. al.

ABSTRACT

Intelligent AI-Driven Job Recommendation and Resume Matching System with Multi-Domain Skill Analysis, Contextual Compatibility Scoring, and Integrated Voice-Enabled Career Assistant

The invention relates to an intelligent AI-driven recruitment and career-assistance system that integrates real-time resume processing, multi-domain skill extraction, contextual compatibility scoring, and voice-enabled interaction within a unified full-stack architecture. The system employs advanced natural language processing, semantic analysis, and a hierarchical skill database to extract and validate skills, experience, and educational attributes from unstructured resume data. A dynamic matching engine computes job-candidate compatibility scores using adaptive weights based on role-specific parameters and industry trends. A responsive frontend interface provides interactive visualizations, instant job recommendations, and detailed gap analyses, while an integrated voice assistant interprets spoken commands to deliver hands-free navigation and career guidance. The backend server ensures synchronized communication between all components, enabling immediate updates and a seamless user experience. The invention offers a comprehensive, context-aware, and continuously improving platform that significantly enhances recruitment efficiency and personalized career mapping.

Accompanied Drawing **[FIGS. 1-4]**

Dated this 01st day of December 2025

Applicant(s)

Satej Mahendra Rajeshirke et. al.